Human Activity Recognition Using Smartphone Accelerometer

Student ID: 24f3100260
Sensors and Applications Laboratory
Week 4 Report

1. Introduction

The goal of this experiment is to use the built-in linear accelerometer of a smartphone to record motion data and to build a simple algorithm that counts steps and recognises the activity being performed. Four activities were recorded: standing still (static), walking on a flat surface, climbing stairs upward, and descending stairs. The exercise gives hands-on experience with sensor data acquisition, signal preprocessing, peak detection and rule-based activity classification, and highlights how the same acceleration signal looks different across everyday movements.

2. Tools Used

- Sensor: smartphone linear accelerometer (gravity component removed at source)
- Data collection app: on-device sensor-logging application exporting linear-accelerometer readings to CSV
- Requested sampling rate: 100 Hz (maximum device sample rate 200 Hz, per the recording metadata)
- Analysis environment: Python 3, using NumPy, Pandas, SciPy (find_peaks) and Matplotlib

Four CSV recordings were collected, one per activity: Static.csv, walking.csv, UP_stairs.csv and Walking_Down_the_Stairs.csv.

3. Basic Idea

Each recording contains the acceleration measured along the X, Y and Z axes, ax, ay and az. These three components are combined into a single resultant magnitude that captures the overall intensity of motion regardless of the phone's orientation:

$$\|a\| = \sqrt{(ax^2 + ay^2 + az^2)}$$

The magnitude signal is smoothed and then searched for peaks: each qualifying peak is treated as one step. Walking, climbing and descending stairs produce visibly different peak heights, spacing and variability, which forms the basis for classifying the activity.

4. Data Collection Summary

- Static: phone held still, ~28.5 s, used as a resting/calibration baseline
- Walking: flat-surface walking, ~89.0 s
- Climbing Up Stairs: ~27.9 s of stair ascent.

- Climbing Down Stairs: ~26.1 s of stair descent.

Total: 4 recordings, roughly 171.5 s and around 17,000 accelerometer samples combined.

5. Algorithm Description

5.1 Magnitude Calculation

For every sample, the 3-axis magnitude is computed as shown above, giving a single scalar time series per recording.

5.2 Smoothing

A Simple Moving Average (SMA) filter with a window of 15 samples (about 0.15 s at 100 Hz) is applied to the magnitude signal to suppress high-frequency sensor noise while preserving the shape of each step.

5.3 Peak Detection

SciPy's find_peaks is used on the smoothed magnitude to locate steps. A minimum peak height of 1.8 m/s² is required for the moving activities (1.0 m/s² for the static recording, where peaks are in any case suppressed), and a minimum spacing of 35 samples (~0.35 s, capping cadence at about 2.8 steps/s) is enforced between consecutive peaks so that a single footfall is not counted twice.

5.4 Rule-Based Classification

Each recording is labelled using simple statistics of its magnitude signal:

- Standard deviation < 0.1 m/s² → Static / Resting
- Mean < 2.2 m/s² and maximum < 8.0 m/s² → Walking (Flat)
- Maximum ≥ 10.0 m/s² or standard deviation > 1.4 m/s² → Stairs (High Impact / Down or Up)
- Otherwise → Stairs (Up)

6. Results

Running the algorithm on the four recordings gave the following step counts and classifications:

Activity	Duration (s)	Data Points	Mean Mag (m/s²)	Std Dev (m/s²)	Max Mag (m/s²)	Steps	Cadence (steps/s)	Classified As
Static	28.53	2835	0.010	0.010	0.374	0	0.00	Static / Resting
Walking	89.00	8843	1.830	0.831	6.168	136	1.53	Walking (Flat)
Climbing Up Stairs	27.90	2773	2.779	1.396	9.713	49	1.76	Stairs (Up)
Climbing Down Stairs	26.09	2593	2.522	1.451	12.048	48	1.84	Stairs (High Impact)

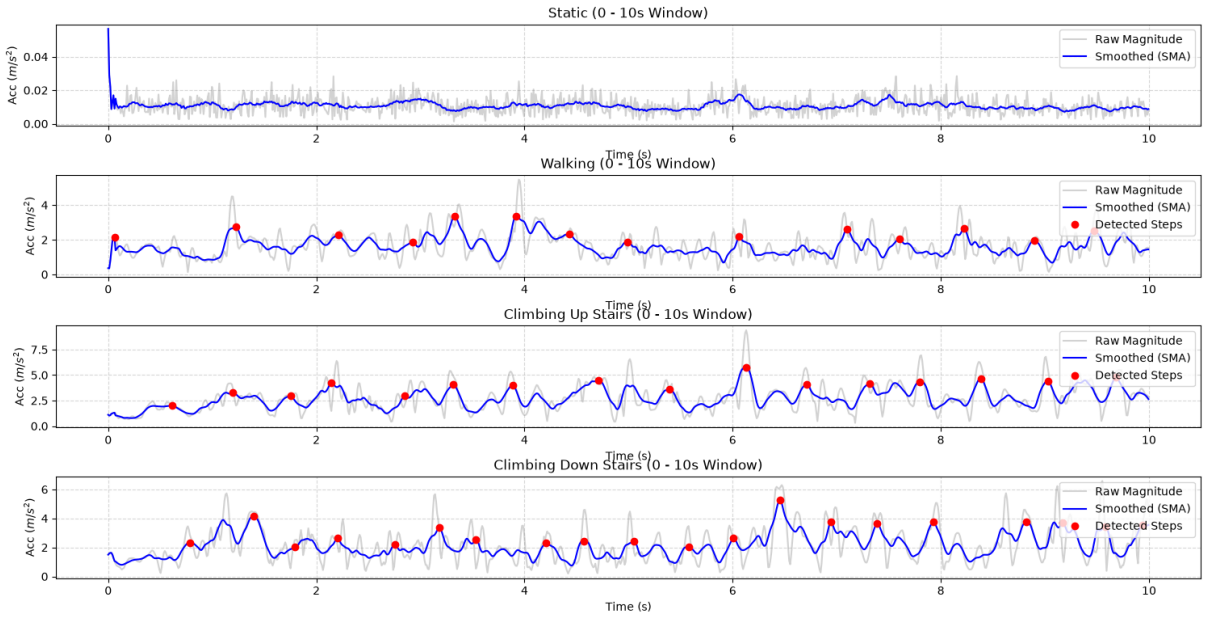


Figure 1: Raw and smoothed acceleration magnitude with detected steps, first 10 s of each activity (Static, Walking, Climbing Up Stairs, Climbing Down Stairs).

7. Discussion

The static recording correctly produced zero detected steps and the lowest standard deviation (0.010 m/s²), confirming that the phone was at rest and that the sensor baseline was clean.

Walking produced 136 steps over 89.0 s, a cadence of 1.53 steps/s, and was classified as flat walking, consistent with its moderate mean magnitude and bounded maximum acceleration.

Both stair activities showed higher mean magnitude and greater variability than walking, as expected for the more effortful, less uniform motion of climbing. Descending stairs produced the highest maximum magnitude (12.048 m/s²) and standard deviation (1.451 m/s²) of all four activities, which is why it was classified as high-impact stair movement; the impact of the foot landing on each step down tends to be sharper than the push-off motion of climbing up. Climbing up stairs fell just under the high-impact threshold on standard deviation but still showed a clearly elevated mean and maximum compared to walking.

This mirrors a common finding in accelerometer-based activity recognition: walking is the easiest activity to separate cleanly, while the two stair activities can sit close to each other and to walking on any single feature, since climbing style, phone placement and orientation all affect the signal. Adding frequency-domain features or combining several axes' statistics (rather than a single magnitude threshold) would likely sharpen the boundary between climbing up and climbing down.

8. Conclusion

A smartphone accelerometer and peak-detection pipeline successfully counted steps and classified basic activities, with, however, stair ascent and descent requiring refined, real-time thresholds. The method proves effective offline, but a machine learning classifier is recommended for improved, robust, and real-time separation of stair activities.